PRANEET SINGH

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RESEARCH FOCUS

At Dolby I develop learned codecs for images, video and 3D scenes, contribute to JVET's next-generation video coding, and build video foundation models. A good codec learns a compact representation of how the world looks and moves, and most video foundation models train and generate in such latent spaces. As these models scale, better compression lets them handle longer, higher-resolution video on fixed compute. Language and speech models show what scale can do; I want to bring compression to video foundation models and world models next.

EDUCATION

Purdue University, West Lafayette, IN *Aug 2019 – May 2025*

PhD, Electrical and Computer Engineering. Advisors: Dr. Amy R. Reibman, Dr. Edward J. Delp

Thesis: Task-Aware Video Compression and Quality Estimation in Practical Video Analytics Systems

PhD research funded by Ford Motor Company

Ramaiah Institute of Technology, Bangalore, India *Jun 2013 – Jun 2017*

B.E., Electronics and Communication

EXPERIENCE

Dolby Laboratories *Jun 2025 – Present*

Sr. Neural Video Processing Engineer, Video Coding Standards (IP Legal). Manager: Dr. Brian Lee

– Learned codecs for images, video and 3D scenes; generative (diffusion, flow-matching) ultra-low-bitrate compression.
– Build video foundation models for neural video processing.
– JVET contributions to Next Generation Video Coding (beyond VVC) on end-to-end learned image/video compression.

Dolby Laboratories *Sep 2024 – Dec 2024*

Video Coding Intern, IP Legal. Manager: Dr. Peng Yin

– Compressed 3D Gaussian Splats by mapping spherical harmonics to ordered 2D frames coded with inter-frame video tools; 12% better efficiency than prior methods (ICME-W 2026, invention disclosure).
– Combined neural networks with conventional codec tools for 3D scene compression.

Dolby Laboratories *May 2023 – Aug 2023*

Imaging & Machine Learning Intern, ATG. Manager: Dr. Guan-Ming Su

– Built NeRVA, one neural field that jointly represents video and audio via shared time-stamp coordinates (ICME 2024 Oral; patent filed).
– Compressed the representation with post-training pruning and quantization.

Apple *May 2022 – Aug 2022*

Video Codec Standards Intern, IMG. Manager: Dr. Alexis Tourapis

– Characterized AV2 common-test-condition sequences by spatial, temporal, noise and compressibility properties (AOM CWG-C047).
– Explored ML-based spatial and temporal super-resolution for scalable video coding.

Purdue University *Aug 2019 – May 2025*

Graduate Research Assistant, VADL Lab; Lead TA, Signals and Systems

– Task-aware video coding and quality estimation for machine vision; Ford-funded, deployed in AV testbeds.
– Built an in-vehicle multi-camera face analytics system and data-collection pipeline for Ford.
– Lead graduate TA for ECE 301 Signals and Systems (Fall 2023, Spring 2024).

Indian Institute of Science *Apr 2018 – Aug 2019*

Research Assistant. Advisors: Dr. Abhay Sharma, Dr. Raghu Krishnapuram

– Traffic analytics framework and dataset for Indian roads from a monocular surveillance camera network: vehicle counting, license-plate detection, speed and queue-length estimation; deployed in Electronic City, Bangalore.
– Real-time web server for live RTMP/HLS video streams with routing, congestion management and load balancing.

Flux Auto *Sep 2017 – Feb 2018*

Self-Driving Engineer

– Lane and pedestrian detection for Level-3 autonomous trucks; over-the-air update infrastructure.

ECI Telecom *Mar 2017 – Nov 2017*

R&D Software Engineer

– Test automation frameworks for optical network systems: add-drop multiplexers and Ethernet Layer 2 functions.

RESEARCH

MixCompress: Mixture of Experts for Variable Rate Learned Image Compression

ECCV 2026. Invention disclosure filed.

– One mixture-of-experts model covers all rates, replacing a bank of single-rate codecs; −20% to −29% BD-rate (PSNR) vs. VVC (VTM-23.1) on Kodak, CLIC and Tecnick.

Optimal Frame Ordering and Inter-frame Compression of Spherical Harmonics for 3DGS Size Reduction

ICME Workshops 2026. Invention disclosure filed.

– Map spherical harmonics to ordered 2D grids so video codecs exploit inter-frame redundancy; 12% higher compression efficiency than prior 3DGS methods.

NeRVA: Joint Implicit Neural Representations for Videos and Audios

ICME 2024 (Oral). Patent filed.

– One neural field jointly encodes synchronized video and audio via shared time-stamp coordinates; compressed by pruning and quantization.

Task-Aware Image Quality Estimators for Face Detection

EURASIP J. Image and Video Processing, 2024

– First unsupervised quality estimator tied to face detection performance, using DropBlock-based robustness; filtering the worst 5% of faces lifts recognition TPR from 0.73 to 0.85 (FPR 10^{-4}).

Face Quality Evaluation at Scale

MMSP 2024; MMSP 2023; Electronic Imaging 2023; MIPR 2022. Funded by Ford.

– Gallery-Query protocol needs 96% fewer comparisons on Glint360k (17M images); stress tests over 33M+ pairs expose a 24% drop in top estimators.

– End-to-end multi-camera face analytics system and dataset; helped Ford reduce captured/stored training data.

Understanding the Characteristics and Compressibility of AOM AV2 CTC Sequences

AOM CWG-C047, 2022. Apple.

– Analyzed AV2 common-test-condition sequences by spatial, temporal, noise and compressibility characteristics.

Video-Analytics Task-Aware Quad-Tree Partitioning and Quantization for HEVC

ICIP 2022. Funded by Ford.

– Task-aware partitioning and rate-distortion optimization in HM: 6% lower bitrate, 15% faster encoding, no loss in detection accuracy; deployed in Ford AV testbeds.

Lightweight Compression of Intermediate Neural Network Features

Purdue, 2021

– Evaluated HEVC and autoencoders for coding intermediate features of split neural networks (video coding for machines).

Edge-Detect: Edge-Centric Network Intrusion Detection using Deep Neural Network

IEEE CCNC 2021

– Lightweight RNN detects DDoS attacks on edge devices; state-of-the-art results on UNSW-NB15.

Animal Localization in Camera-Trap Images with Complex Backgrounds

IEEE SSIAI 2020

– Unsupervised robust-PCA saliency separates animals from complex backgrounds; matches learned models such as R3-Net without training.

MV-Tractus: Fast Motion Vector Extraction from H.264 Streams

Zenodo, 2018. 124 GitHub stars.

– Open-source tool that extracts motion vectors directly from H.264 bitstreams.

PUBLICATIONS

- *MixCompress: Mixture of Experts for Variable Rate Learned Image Compression.* ECCV, 2026.
- *Optimal Frame Ordering and Inter-frame Compression of Spherical Harmonics for 3D Gaussian Splatting Size Reduction.* IEEE ICME Workshops, 2026.
- *Image Quality Assessment in End-to-end Face Analytics Systems.* IEEE MMSP, 2024.
- *Task-aware Image Quality Estimators for Face Detection.* EURASIP J. Image and Video Processing, 2024.
- *NeRVA: Joint Implicit Neural Representations for Videos and Audios.* IEEE ICME (Oral), 2024.
- *Gallery-Query Protocol for Evaluating Face Image Quality Metrics.* IEEE MMSP, 2023.
- *End-to-end Evaluation of Practical Video Analytics Systems for Face Detection and Recognition.* Electronic Imaging (AVM), 2023.
- *Video-Analytics Task-Aware Quad-Tree Partitioning and Quantization for HEVC.* IEEE ICIP, 2022.
- *Evaluating Image Quality Estimators for Face Matching.* IEEE MIPR, 2022.
- *Understanding the Characteristics and Compressibility of AOM AV2 CTC Sequences.* AOM CWG-C047, 2022.
- *Edge-Detect: Edge-Centric Network Intrusion Detection using Deep Neural Network.* IEEE CCNC, 2021.
- *Animal Localization in Camera-Trap Images with Complex Backgrounds.* IEEE SSIAI, 2020.
- *Detection of Anomalous Behaviour in Crowds Using Newton Pratt's Curve Fitting Technique.* ICCCNT, 2018.
- *MV-Tractus: Fast Motion Vector Extraction from H.264 Streams.* Zenodo, 2018 (124 GitHub stars).
- *Performance Evaluation of Cryptographic Ciphers on IoT Devices.* arXiv:1812.02220, 2018.